LAUNCH CANADA
2026 CHALLENGE



Preliminary Design Report for Project *Polaris*

Competing Team 22

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Abstract

McGill Polaris Flight (MPF) is a new group spun-off from the McGill Rocket Team (MRT), with its focus on development of control systems. Project *Polaris* is the first rocket of the team. Project *Polaris* consists of two primary SRAD subsystems: the roll control system, and the avionics, which includes a radio frequency direction of arrival system (RF-DoA). They are integrated onto COTS airframe, recovery system, and solid motor. The avionics consists of SRAD flight computers running SRAD software, with COTS backup. The control system controls the rocket's roll attitude with SRAD controller and mechanisms. To demonstrate the capability of the control system, avionics will determine the azimuthal angle of the ground station using its transmitted RF-signal relative to the rocket frame such that the control system could point the rocket's on-board camera towards the ground station.

The primary technical risks affecting project success are the uncertainties on the new control system design. These risks are mitigated through extensive testing—including component testing and software-in-the-loop (SIL) testing.

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Abbreviations

AGL	Above ground level
CAD	Computer-aided design
CG	Centre of gravity
CONOPS	Concept of operations
COTS	Commercial-off-the-shelf
CP	Centre of pressure
DOF	Degrees of Freedom
DTEG	Launch Canada Design, Test, and Evaluation Guide
FCs	Flight computers
FEA	Finite element analysis
FS	Factor of safety
GS	Ground station
ID	Inner diameter
IMU	Inertial measurement unit
LC	Launch Canada
MCC	Mission control center
MIAE	McGill Institute for Aerospace Engineering
MPF	McGill Polaris Flight
OD	Outer diameter
PCB	Printed circuit board
PDR	Preliminary design review or Preliminary design report
PID	Proportional-integral-derivative
SRAD	Student-researched and designed

Units

A	Ampere
Ah	Ampere-hour
BC	Body caliber
bit/s	Bits per second
ºC	degrees Celsius
cm	Centimeter
dB	Decibel
dBi	Decibel-isotropic
dBm	Decibel-milliwatt
ft	Feet
g	Gravitational force equivalent
GPa	Gigapascal
Hz	Hertz
in	Inch
kHz	Kilohertz
lbf	pound-force
mA	Milliampere
mAh	Milliampere hour
MHz	Megahertz
MPa	Megapascal
kg	Kilogram
km	Kilometer
kN	Kilonewton
kPa	Kilopascal
lbf	Pound-force
lbm	Pound-mass
m	Meter
mm	Millimeter
ms	Millisecond
N	Newton
psi	pounds per square inch
s	Second
V	Volt
W	Watt
Wh	Watt-hour

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1. Introduction

McGill Polaris Flight (MPF) is a new group spun-off from the McGill Rocket Team (MRT), with its focus on development of stability augmentation systems and radio direction finding technology. Flight controls is a discipline crucial to any aerospace vehicle but largely unreachable from traditional amateur rocketry. We want to explore this technology and the possibility of incorporating it onto rockets. We aim to launch a modified LOC Precision IRIS 4" kit, with custom avionics and active roll stability augmentation, in the Technology Development category at Launch Canada 2026. The rocket will consist of an all COTS airframe, recovery package, and solid motor (AeroTech K535W-14A), except the necessary modifications. The avionics will be SRAD, with a COTS flight computer as redundancy.

1.1. Team Organization

The team is supported by McGill Institute of Aerospace Engineering (MIAE). The team reports to the head of MIAE, Prof. Siva Nadarajah. Benedetto D'Agnillo, the Team Captain, is in charge of avionics. Pierre Arbaji, the Chief Engineer, takes care of much of the control system design. Michael Zhou, the Chief Safety Officer, helps with administrative work. The team would also like to thank Majd Boulos for working on early design and testing, and Mattéo Hapiot for conducting in-depth research and development on rocket flight dynamics and control systems.

1.2. Timeline

The timeline for project *Polaris*is periodically refined to reflect the latest developments. Currently, most of the mechanical and electrical design is complete. The team aim to complete design by end of April and complete all components manufacturing by end of May. The team is exploring chances to launch small scale test flights in May and June. Should this be deemed possible, the controller design for both competition rocket and small scale test rocket will be finalized by end of April, and kept being refined based on flight test results.

2. Business Case Analysis

2.1. Research Goals and Value Proposition

Control systems are a less common feature in high power rocketry due to their added complexity and high risk. Avionics systems tend to plateau in terms of research and design year after year designing controllers to simply actuate valves and eject at the right time. With the objective of advancing research and design into state estimation solutions for high-power rocket roll stabilization and control, we believe this to be one of the next big advancements in the McGill rocketry sphere and the Launch Canada community.

We propose a low-cost roll control and RF direction finding solution novel to rocketry, designed for a commercial off-the-shelf rocket airframe.

A canard-based roll authority system provides an accessible entry point into active control without increasing propulsion complexity, as would be required with thrust vectoring.

State estimation is often an overlooked aspect of avionics design. Parachute deployment can be achieved using relatively simple algorithms based on raw acceleration and barometric data, while even more advanced systems tend to only scratch the surface of sensor fusion and filtering techniques. In contrast, an active roll control system requires a robust state estimator, which is why these research and design objectives complement each other particularly well.

Additionally, in an environment where GPS-denied navigation techniques are becoming increasingly relevant, our system employs an RF pseudo-Doppler direction-finding method to track the ground station's position during flight, as well as to enable the ground station to track the rocket. This demonstrates the foundational principles of triangulation and navigation techniques commonly used in the amateur radio community.

2.2. Market Research

The primary market for this project is within student and amateur rocketry communities, where teams increasingly seek to incorporate advanced control and navigation capabilities while remaining constrained by cost, complexity, and development risk. Most high-power amateur rockets rely on passive stabilization and minimal avionics logic, leaving active roll control and state estimation largely unexplored at the student level.

A low-cost, canard-based roll control system paired with robust state estimation provides a practical pathway for teams to advance beyond conventional avionics without introducing significant propulsion complexity. In addition, the demonstration of RF-based, GPS-denied tracking techniques addresses a growing area of interest within both the amateur radio and experimental aerospace communities.

Target users include:

- University and collegiate rocketry teams
- Launch Canada and similar competition teams
- Advanced amateur high-power rocketry groups
- Student researchers in guidance, navigation, and control
- Amateur radio operators interested in airborne direction finding

2.3. Financial Analysis

The following tables detail the cost breakdown to date for the development of the rocket and associated ground station, as well as the breakdown of travel expenses.

Table 2.1 Rocket Cost Summary

Category	Budget ($)	Actual (current) ($)	Variance ($)
Avionics Hardware	500	300	-200
Mechanical Components	200	0	-200
Solid motor	300	293.54	-6.46
COTS airframe	500	439.99	-60.1
Miscellaneous	500	0	-500
Total	**2,000**	**1,033.53**	**-966.56**

Table 2.2 Estimated Travel Expenses for a One-Week Trip from Montréal to Timmins, Ontario

Category	Assumptions	Estimated Cost ($)
Fuel	$\sim$1,100 km round trip, average vehicle	250
Food	7 days at $50/day	350
Accommodation	6 nights at $120/night	720
Local Fuel Costs	Local driving	120
Supplies & Miscellaneous	Contingencies, consumables	100
Total		**1,540**

3. Mission Concept of Operations Overview

The mission concept of operations (CONOPS) for Project *Polaris* is made of ten stages. An overview of the CONOPS is illustrated in Fig. 3.1. Note that the following section is drafted based on the team members' past experiences on hybrid launch operations at LC with the MRT. We are aware that the safety procedures for solid launches and tech devs may be different, and are therefore prepared to adjust our plans accordingly. Note that the flight dynamic values shown here are for the original COTS rocket without control system. The analysis of trajectory with control system activated is ongoing.

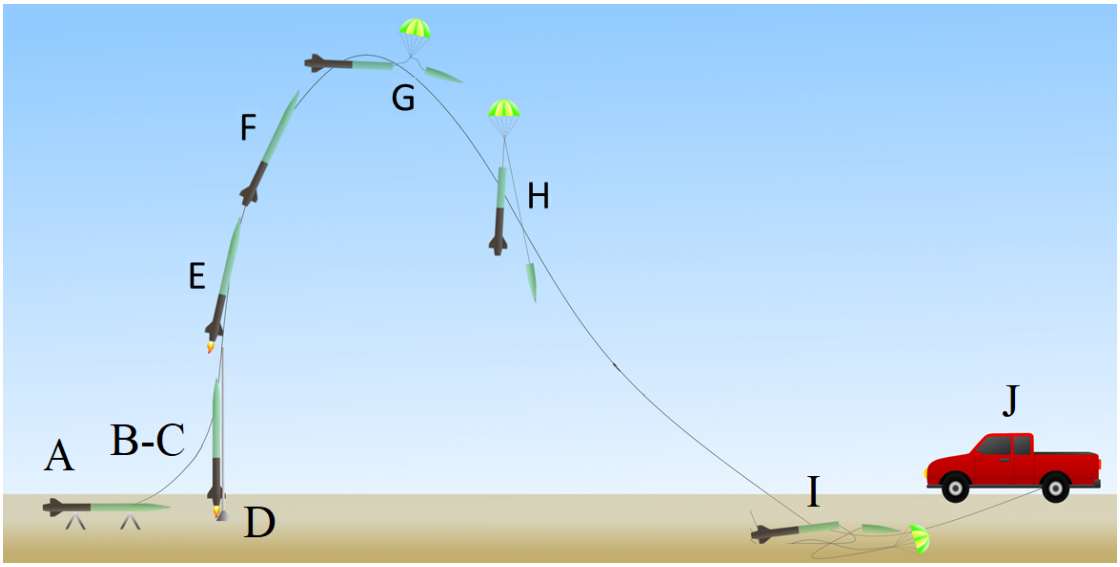


Fig. 3.1 CONOPS overview.

A. Rocket Assembly

Estimated duration (nominal): 30 minutes
Estimated duration (off-nominal): $>$ 1 hour

1. **Start condition**: The base camp is open and the rocket is ready to be assembled.
2. The rocket will be assembled and the control system thoroughly tested to ensure nominal functionalities.
3. Once assembled, the rocket will await flight approval at the assembly area. If not already done, the flight card will be completed at this point, notably the apogee prediction section.
4. **End condition**: The rocket is fully assembled and approved for flight.

B. Rail Loading

Estimated duration (nominal): 15 minutes
Estimated duration (off-nominal): $>$ 30 minutes

1. **Start condition**: The rocket is fully assembled and approved for flight.
2. Once the pad is open, the rocket will be transported to the launch pad. The rocket will then be loaded onto the launch rail.

3. All non-critical personnel will leave the pad, and pre-flight checks will be performed. At this point, members of other teams can still be at the pad. Critical personnel will include, at the very least, one person in charge of avionics and telemetry.

4. At this instant, the ground station will verify that they have established communication with the rocket.

    (a) If communication cannot be established, the rocket will be returned to the assembly area per the approval of LC pad safety officer, where it will be troubleshooted.

5. The control system will be tested to ensure nominal functionalities. In the case where an issue can not be resolved, the rocket will be returned and troubleshooted.

6. The engine igniter will be inserted inside the combustion chamber, and the control system will be armed but locked.

7. All personnel at the pad will evacuate. Avionics will be active and collecting data. The control system will remain on standby.

8. **End condition**: The rocket is in a launch-ready position on the launch rail.

C. Ignition

Estimated duration (nominal): 1 s
Estimated duration (off-nominal): unknown

1. **Start condition**: The rocket was cleared for ignition and takeoff.

2. LC mission control will execute the ignition. If successful, the rocket should begin liftoff.

3. **End condition**: The rocket begins accelerating.

D. Liftoff

Estimated duration (nominal): Less than 1 s.
Estimated duration (off-nominal): N/A

1. **Start condition**: The rocket is accelerating and the engine reaches peak thrust.

    (a) If the structural integrity of the control surfaces is lost from this instant, it may cause the rocket to deviate from the nominal flight path. This outcome will be considered to be catastrophic at take-off (CATO).

2. **End condition**: The first launch lug clears the launch rail.

E. Powered Flight

Estimated duration: 2.9 s

1. **Start condition**: The first launch lug has cleared the launch rail.

    (a) The off-rail velocity depends on the length of rail provided by LC.

2. The engine will continue to accelerate the rocket, for a total burn time of $\approx$ 2.9 s. The avionics will continue acquiring data. The control system will be locked in a neutral state.

(a) If the safety lock fails, the control system will cause the rocket to deviate from the nominal flight path. This outcome will be a loss of control.

3. **End condition**: The engine burns out and the flight enters coasting phase.

F. Coasting Ascent and Control Activation

Estimated duration: 12 s

1. **Start condition**: The engine has burned out.
2. Once the avionics detects the burn out, the control lock will be disabled, and controller will command control surface movements to orient the rocket into a pre-planned roll attitude.
   (a) If avionics fails to detect the burn out, the rocket will continue safe flight without active control. However, the mission will be considered a failure.
3. Once avionics detects the rocket's attitude exceeds 30° from its launch elevation, the control lock will be re-enabled.
   (a) If the control lock fails to be enabled and control system remains activated, the rocket may deviated from a nominal flight path.
4. The rocket will reach an apogee $\approx$ 4400 ft (1341 m).
5. **End condition**: The rocket is about to reach apogee.

G. Ejection

Estimated duration: 1 s

1. **Start condition**: The flight duration reaches the preset ejection time.
2. The ejection is triggered by a preset time-delay ejection charge on the engine.
   (a) The ejection may occur a short time before or after reaching the actual apogee. This is nominal.
   (b) If the ejection occurs too early, the harness will be tangled, resulting in a ballistic descent.
   (c) If the ejection occurs too late, the harness might not withstand the shock, resulting in a ballistic descent.
3. Ejection will occur the rocket body will separate in two sections, connected by harnesses. The two sections will continue moving away from the body until the harness reach line stretch.
4. **End condition**: The rocket is separated from ejection, and the harness reaches line stretch.

H. Main Descent

Estimated duration (nominal): 160 s
Estimated duration (off-nominal): N/A if there is failure.

1. **Start condition**: Successful ejection.
2. If the main is successfully deployed at apogee, the rocket will proceed with a controlled descent under main for $approx$ 160 s until it impacts the ground with a terminal speed of 8.05 m/s.
3. **End condition**: The rocket is about to impact the ground.

I. Touchdown

Estimated duration: 1 s

1. **Start condition**: The rocket has made initial contact with the ground.
   (a) If the separated bodies are no longer connected due to recovery harness failure, the two parts will have landed at different locations.
   (b) Avionics telemetry will remain powered on (with an energy saving profile) for as long as possible until recovery is successful.
2. **End condition**: The rocket and all its deployed contents come to rest on the ground.

J. Ground Recovery

Estimated duration: up to hours or days

1. **Start condition**: A recovery team is gathered together and receives permission from the MCC to go search for the rocket.
2. Due to the wide range of possible scenarios, recovery operations will not be discussed here.
3. **End condition**: The recovery team has returned to the MCC, either with or without the rocket body for post-flight evaluation.

4. Preliminary System Architecture Overview

4.1. System description

The system is composed of (i) a modified air vehicle built from a LOC Precision IRIS kit airframe and recovery system, (ii) an active roll-control subsystem implemented using a pair of forward-mounted canards actuated by two independent servos, and (iii) an onboard avionics subsystem for sensing, estimation, control computation, and logging.

In contrast to a fin-mounted control-surface approach, the present architecture adopts a canard-based actuation layout where control surfaces are placed near the forward section of the rocket, enabling aerodynamic control authority through asymmetrical lift generated ahead of the main fin set.

Table 4.1 System parameters.

Parameter	Value
Rocket kit	LOC Precision Yank IRIS4 (IRIS 4 in)
Diameter	4 in
Height	69 in
Listed weight	5.5 lb
Motor mount	54 mm
Motor	AeroTech K535W-14A (54 mm $\times$ 358 mm, 1434 N·s, 2.8 s burn)
Roll-control concept	Forward canards (2 surfaces)
Number of actuators	2 servos (one per canard)
Primary feedback sensor	IMU gyroscope (roll rate)

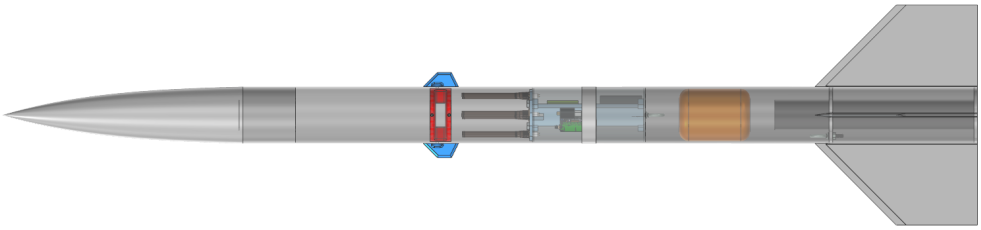


Fig. 4.1 CAD overview of the flight-test rocket architecture.

4.2. Control Mechanism Overview

This subsection describes the mechanical architecture of the canard-based control system integrated into the baseline airframe. The mechanism consists of two externally mounted canards that are mechanically linked through a bevel gear transmission and actuated by a single high-torque servo. The design ensures synchronized motion of both control surfaces while incorporating a passive mechanical safety feature to prevent excessive deflection.

4.2.1. Mechanical layout

The control mechanism is implemented as a self-contained structural module intended to be primarily manufactured using additive manufacturing. The module houses a single centrally mounted servo and an internal bevel gear transmission that mechanically links the two canards. Integrated wiring channels allow servo leads to be routed internally through the airframe, minimizing external exposure and simplifying assembly.

Each canard is mounted externally on the airframe and interfaces with the internal mechanism through a dedicated shaft. Internally, these shafts are coupled to secondary bevel gears, each meshing with a central bevel gear driven directly by the servo output shaft. Rotation of the central bevel gear therefore produces synchronized and symmetric deflection of both canards, eliminating the possibility of differential deflection due to actuator mismatch.

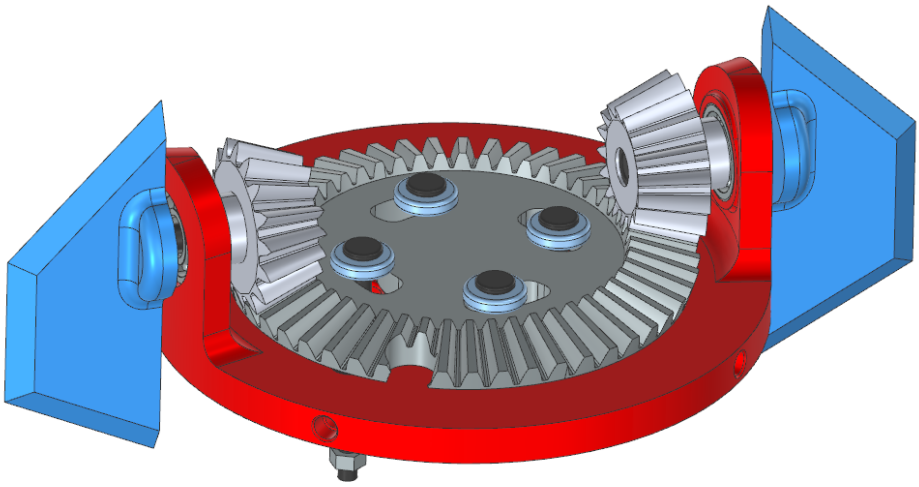


Fig. 4.2 Canard control mechanism during nominal operation.

4.2.2. Over-rotation safety lock

To prevent excessive canard deflection and protect the mechanism from mechanical overload or unsafe aerodynamic conditions, a passive over-rotation safety lock is integrated into the central gear assembly. The allowable angular travel of the mechanism is conservatively limited to ensure operation remains well within a safe structural and aerodynamic envelope.

The central bevel gear contains a set of precisely positioned radial holes corresponding to the maximum permissible rotation angle. Beneath the gear, spring-loaded locking rods are installed within the module housing and are continuously biased upward toward the gear. During nominal operation, the solid portions of the gear maintain the rods in a compressed state.

If the servo drives the mechanism beyond the predefined angular limit, one of the holes in the central bevel gear aligns with a corresponding locking rod. The rod is then driven upward by the spring force into the hole, mechanically engaging the gear and physically preventing any further rotation. Once engaged, the lock acts as a hard mechanical stop, independent of software or servo control, thereby providing a fail-safe protection against over-deflection.

4.2.3. Return-to-neutral safety lock

The return-to-neutral safety lock is still being designed. The preliminary idea centers on a clutch type mechanism that connects the actuator to the bevel gear. Should any unsafe scenario occur, it will trigger the clutch to disengage. A restoring spring on the rack will return it to neutral position.

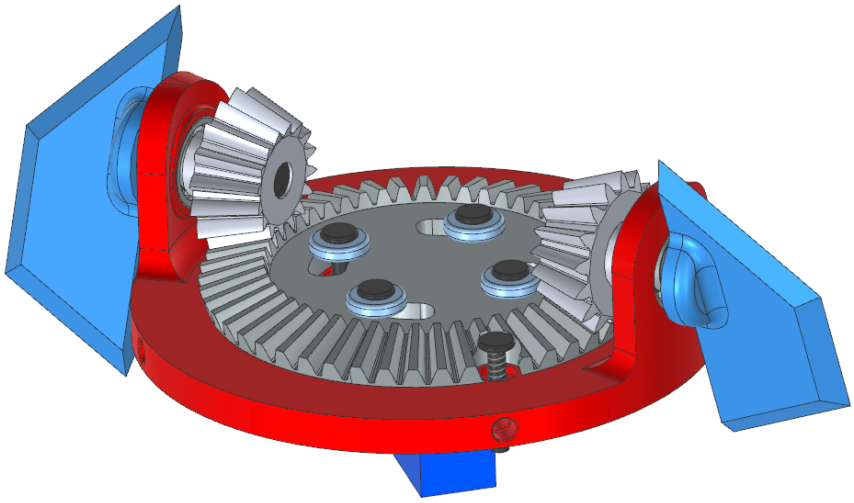


Fig. 4.3 Canard control mechanism after activation of the over-rotation safety lock.

4.2.4. Servo torque sizing

Servo selection is driven by the maximum aerodynamic hinge moment on each control surface at the highest expected dynamic pressure. Using a constant, uniform-velocity assumption, the torque required to hold a surface at a given deflection can be approximated as

$$T = \frac{1}{2}\rho V^2 A C_L L, \tag{4.1}$$

where T is torque, L is the lever arm, ρ is air density (sea-level), V is free-stream velocity, A is control-surface planform area, and C_L is an effective coefficient. For bounded deflections, C_L is approximated as

$$C_L \approx \sin(2\delta)\,, \tag{4.2}$$

with $\delta \leq \delta_{\max}$.

For the present design, the maximum expected velocity is taken as

$$V_{\max} = 768\ \text{ft/s} = 234.09\ \text{m/s}, \tag{4.3}$$

and the control surface area is approximated as a right triangle of base 3 in and height 1.25 in:

$$A = \frac{1}{2}\,(3\ \text{in})\,(1.25\ \text{in}) = 1.875\ \text{in}^2 = 1.2097 \times 10^{-3}\ \text{m}^2. \tag{4.4}$$

The canard chord is $c = 1.25$ in, and the aerodynamic moment arm is approximated using the aerodynamic center at $0.25c$:

$$L = 0.25c = 0.3125\ \text{in} = 7.94 \times 10^{-3}\ \text{m}. \tag{4.5}$$

Taking the maximum deflection as in the previous report, $\delta_{\max} = 20^\circ$, the required torque evaluates to

$$T_{\text{req}} = \frac{1}{2}(1.225)\,(234.09)^2\,(1.2097 \times 10^{-3})\,\sin(40^\circ)\,(7.94 \times 10^{-3}) = 0.207\ \text{N}\cdot\text{m}. \tag{4.6}$$

Applying a factor of safety of 2 yields the minimum required stall torque:

$$T_{\min} = 2T_{\text{req}} = 0.414\ \text{N}\cdot\text{m} \approx 4.22\ \text{kgf}\cdot\text{cm}. \tag{4.7}$$

5. Requirements and Design Goals

To avoid bloating the report or merely copy and paste every item from the DTEG, in this section we focus on the mission of project *Polaris* and the requirments pertaining to SRAD avionics and control system.

5.1. Mission Requirements

5.1.1. Functional Mission Requirements

Subteam	Specification	REQ ID/Source	Chosen Design Description	Planned way to show Compliance
CS	Launch vehicle active flight control systems shall be optionally implemented strictly for roll stability augmentation or aerodynamic ”braking”.	LC DTEG R4 R7,1.1	Roll control system is implemented to stabilize roll rate and point the camera to the ground station	No pitch or yaw control capability.
CS	Under no circumstances shall a launch vehicle entered in the LC Challenge be actively guided towards a designated spatial target.	LC DTEG R4 R7,1.2	No navigation or guidance capability	N/A
CS	The flight shall have a pre-determined purpose to demonstrate the capability of the control system.	MPF	Radio Frequency Direction of Arrival System	See section 7.
CS	The flight shall have a short boosting phase and long coasting phase such that there is sufficient time for control system activation.	MPF	The selected airframe and motor provides more than 10 seconds of coasting	See section 6.
AV	All rockets at the Launch Canada challenge shall incorporate a GPS tracking solution.	LC DTEG R4 R6.4.1	Both the SRAD system and COTS altimeter have GPS tracking and telemetry to transmit the location	Perform multiple cold start GPS lock validation tests in different environments
AV	The rockets shall carry a COTS barometric altimeter with on-board data storage, which will provide and official log of apogee for scoring.	LC R&R Sec 3.3	The Telemega 7.0 is used as COTS altimeter	Perform vacuum bot barometer tests and ensure correct altitude logging.

5.1.2. Performance Mission Requirements

Subteam	Specification	REQ ID/Source	Chosen Design Description	Planned way to show Compliance
CS	Under no circumstances shall the first test of an actively controlled launch vehicle exceed an impulse of 5 kN·s (Solid Motors) or 10 kN·s (Hybrid or Liquid).	LC DTEG R4 R7.1.3	COTS solid motor AeroTech K535W-14A is selected.	The motor has total impulse of 1434 Ns.

5.2. System Requirements

5.2.1. Functional System Requirements

Subteam	Specification	REQ ID/Source	Chosen Design Description	Planned way to show Compliance
CS	CAS or ACS shall mechanically lock in a neutral state whenever either an abort signal is received for any reason, primary system power is lost, or the launch vehicle's attitude exceeds 30° from its launch elevation. Any one of these conditions being met shall trigger the fail-safe, neutral system state.	LC DTEG R4 R7.3.1	Mechanical safe lock (sec 4.2.), and flight computer logic	Ground testing
CS	CAS or ACS shall mechanically lock in a neutral state until the mission's boost phase has ended (i.e., all propulsive stages have ceased producing thrust).	LC DTEG R4 R7.4	Mechanical safe lock (4.2.), and flight computer logic	Ground testing
AV	All onboard electrical systems, including avionics, global positioning system (GPS) and telemetry, shall be tested as an integrated system to ensure that no components cause any apparent interference with any other.	LC DTEG R4 R6.1.2	Telemetry antennas will be placed far from GPS antennas, and the duplex radio system works such that RF-DoA and telemetry periods do not overload any receivers.	Perform telemetry range tests while ensuring a GPS lock.

AV	Antennas shall be located in sections of the vehicle that do not significantly attenuate the signal. Notably, carbon fiber and metal will block radio frequency (RF) signals. Fiberglass is preferred for RF transparency.	LC DTEG R4 R6.3.2	The airframe is primarily made of cardboard and wood, with some fiberglass reinforced sections.	Perform a range test with the system inside the rocket airframe.
AV	The SRAD control system hardware and algorithm shall be validated using a software in the loop (SIL) test.	MPF	A comprehensive physics based model with fin deflection/servo feedback will be developed for simulation.	The model will be used to simulate the rocket flight with the tuned control system to validate it.

5.2.2. Technical System Requirements

Subteam	Specification	REQ ID/Source	Chosen Design Description	Planned way to show Compliance
CS	Launch vehicles implementing active flight controls shall be naturally stable without these controls being implemented (e.g., 2 launch vehicles implementing active flight controls shall be naturally stable without these controls being implemented (e.g., the launch vehicle may be flown with the control actuator system (CAS) – including any control surfaces – either removed or rendered inert and mechanically locked, without becoming unstable during ascent).	LC DTEG R4 R7.2.1	COTS airframe LOC Yank Iris 4" is selected.	OR simulations.

5.3. Subsystem Requirements

5.3.1. Functional Subsystem Requirements

Subteam	Specification	REQ ID/Source	Chosen Design Description	Planned way to show Compliance
CS	The active control system shall implement a canard-type control surface located above the engine section of the rocket.	MPF	As described.	N/A
CS	The actuators shall withstand the maximum torque the control surfaces may require during flight.	MPF	TBD	Calculations and high factor of safety
CS	The controller must be sufficiently robust to address the uncertainties in flight conditions and disturbances.	MPF	See section 9.	Simulations.
AV	All safety critical wiring shall implement a cable management solution (e.g., wire ties, wiring harnesses, cable raceways) which will prevent tangling and excessive free of significant wiring/cable lengths due to expected launch or other loads.	LC DTEG R4 R6.2.1	All cable routing will be accounted for in the mechanical design of the AV bay by ensuring there are proper zip-tie attachments for wire harnesses as well as general cable routing to ease assembly and minimize tangling.	All wire connections shall endure a rough pull test.
AV	All wiring and cables shall include enough slack at all connections/terminals to prevent unintentional de-mating due to expected launch loads transferred into wiring/cables at physical interfaces.	LC DTEG R4 R6.2.2	Same as above	Same as above
AV	Any state detection system shall be tested and tuned prior to arrival at competition and checked after pad setup in case anything was damaged, or calibration was lost in transit.	LC DTEG R4 R2.5.33	The control system startup will perform a self validation step using Hall effect sensors to ensure the mechanism functions as intended.	The validation step will be tested to accurately detect any off nominal cases at every startup or manual command.

AV	The SRAD avionics shall autonomously detect off-nominal states and return the control surfaces to their neutral state.	MPF	The controller's estimator will detect if the rocket's angle is in an off-nominal state based on IMU data or if the estimator's bias becomes too large to perform maneuvers confidently. This will disengage the clutch to the control system and engage the safety lock.	The rocket will be manipulated manually to mimic off nominal states in order to validate the safety lock.
AV	The ground station control system shall have a manual emergency stop button to return the control surfaces to their neutral state.	MPF	The ground station controller is connected to a mechanical emergency stop button that sends the telemetry command to disable all controls and engage the safety lock.	Test that the emergency stop command and button work reliably together.

5.3.2. Technical Subsystem Requirements

Subteam	**Specification**	**REQ ID/Source**	**Chosen Design Description**	**Planned way to show Compliance**
CS	The control surfaces shall be placed at a height as close to the rocket CG as practical to minimize inadvertent pitch or yaw moment.	MPF	The control mechanism is placed immediately above the AV bay.	See 4.
AV	The SRAD direction finding system shall reliably estimate the position of the ground RF source reliably within 5 degrees uncertainty.	MPF	The DoA algorithm and physical system is designed to measure the angle to better than 5 degrees uncertainty.	The DoA system will be tested in large fields and during a long range test.

6. Performance Analysis

6.1. Flight performance

To support preliminary sizing and feasibility checks, the baseline rocket configuration is modeled in OpenRocket to obtain trajectory predictions, including maximum velocity, off-rail conditions, and predicted apogee.

The simulation launch conditions are chosen to approximate Launch Canada operating conditions by using a representative launch-site location (coordinates set to the Launch Canada competition area in Timmins, Ontario) and typical competition-period winds.

Table 6.1 Flight performance parameters (OpenRocket prediction).

Description	Value
Dry mass	3.5 kg
Liftoff (wet) mass	4.76 kg
Peak thrust	655 N
Burn time	2.8 s
Total impulse	1429 N·s
Minimum static margin during ascent	3.38 BC
Maximum velocity	770 ft/s
Maximum acceleration	411 ft/s^2
Predicted apogee	4558 ft

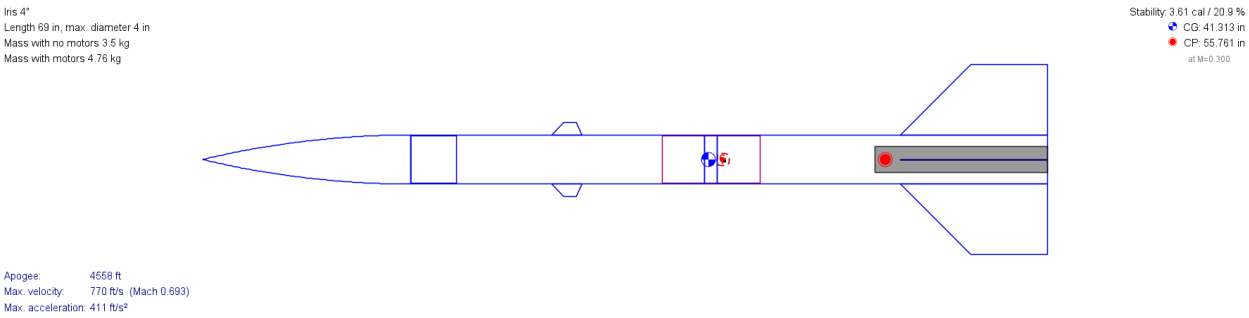


Fig. 6.1 OpenRocket model setup used for baseline trajectory prediction.

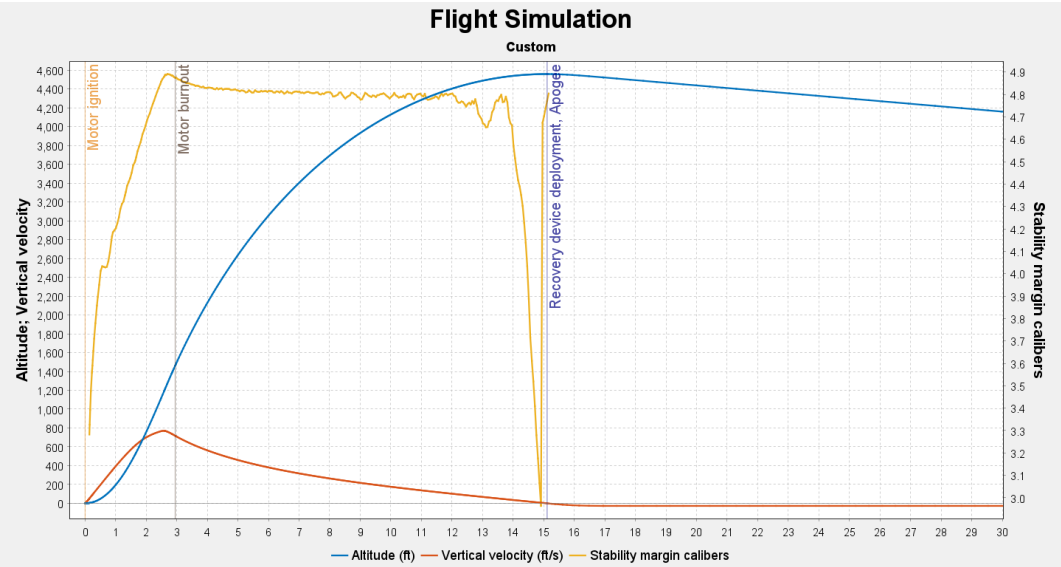


Fig. 6.2 Predicted flight altitude and stability margin from OpenRocket.

7. Radio Frequency Direction of Arrival (RF-DoA) System

7.1. Motivation

In order to demonstrate the roll control capability on *Polaris*, we decided to design an RF-based direction of arrival estimation system which can determine the azimuthal angle of the ground station using its transmitted RF-signal relative to the rocket frame such that the control system could point the rocket's on-board camera towards us (the ground station operators) and take a high-power selfie.

7.2. Background

In more conventional direction of arrival methods, a multi-channel coherent software-defined radio (SDR) is used such that measuring the phase difference received on each channel determines the estimated direction of arrival of a signal. This is essentially the inverse of a phased array used for receiving instead of transmitting in a certain direction.

While these systems work incredibly well and are used in many real-world communications applications, the entry-level price can be very steep for multi-channel coherent SDRs.

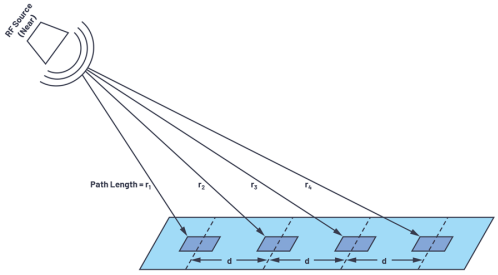


Fig. 7.1 Direction of arrival for multi-channel phased array (source: Analog Devices)

7.3. Pseudo-Doppler Direction of Arrival (PD-DoA)

In the world of amateur radio, there is a well-known technique called pseudo-Doppler DoA. Assume an antenna is rotating at a high angular rate around in a circle while it receives a signal from a transmitting source, then when the antenna is moving towards the source, it should see the signal's frequency increase, and decrease while it moves away from the source due to the Doppler effect.

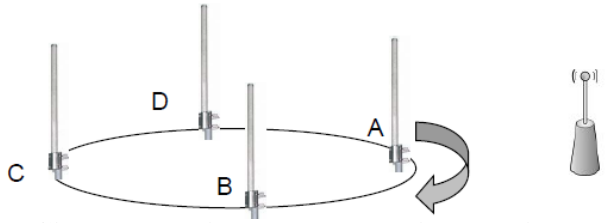


Fig. 7.2 Circular rotating antenna relative to a signal source.

Better angular resolution requires rotating at increasingly faster speeds, which isn't feasible for a mechanical system. This can be emulated by switching at the same rate between a series of stationary antennas placed in a circular array. This is the pseudo-Doppler method.

7.4. Measuring direction of arrival

The DoA system tracks a sine-wave carrier signal produced at the ground station at a frequency f_c using a 6 antenna circular array equally spaced at a radius r. The angular position of antenna n is $\phi_n = 2\pi n/6$. Assume the incoming plane wave arrives from an azimuthal angle θ.

The plane wave arriving from direction θ travels an extra path length at antenna n given by:

$$\Delta L_n = r\cos(\theta - \phi_n)$$

This corresponds to a phase offset:

$$\Delta\varphi_n = \frac{2\pi}{\lambda} r\cos(\theta - \phi_n)$$

Then the received signal at antenna n is:

$$s_n(t) = A\cos(2\pi f_c t + \Delta\varphi_n)$$

This represents what the phased array measures simultaneously. When switching between antennas at a frequency f_s (angular switching speed $\omega = 2\pi f_s$), we are effectively time-multiplexing the antennas.

Define the continuous angular position (before 6 antenna discretization):

$$\phi(t) = \omega t$$

The phase becomes a function of time:

$$\Delta\phi(t) = \frac{2\pi r}{\lambda}\cos(\theta - \omega t)$$

Received signal becomes:

$$s(t) = A\cos(2\pi f_c t + \beta\cos(\theta - \omega t))$$

where $\beta = 2\pi r/\lambda$. This is an angle-modulated signal at frequency f_s (FM).

A physically rotating antenna would have radial velocity $v_r(t) = \omega r\sin(\theta - \omega t)$. The true Doppler shift is then:

$$\Delta f(t) = \frac{v_r(t)}{\lambda} = \frac{\omega r}{\lambda}\sin(\theta - \omega t)$$

Sampling this continuous function at equal time intervals between antennas at a fixed rate is where this becomes pseudo-Doppler.

7.5. Error/precision estimation

For a sinusoid in white noise, the Cramér-Rao bound gives the error on the Doppler sinusoid as:

$$\sigma_\phi^2 \geq \frac{2}{\mathrm{SNR}\cdot M} \approx \sigma_\theta^2$$

where M is the number of independent samples (cycles).

$$\implies \sigma_\theta \approx \sqrt{\frac{2}{\mathrm{SNR}\cdot \mathrm{M}}}$$

The Doppler SNR depends on:

$$\mathrm{SNR}_{\mathrm{Doppler}} \propto \left(\frac{\omega r}{\lambda}^2\right)\cdot \mathrm{SNR}_{\mathrm{carrier}}$$

For a carrier SNR of $12\mathrm{dB} \approx 3.16$ and an obervation window of 1000 cycles (≈ 0.1s for 10kHz cycle frequency), gives a theoretical best case for angular resolution of:

$$\sigma_\theta \approx \sqrt{\frac{2}{15.8\cdot 1000}} \approx 0.0113\mathrm{rad} \approx 0.65^\circ$$

In real-world conditions considering antenna mismatch, switching transients, multipath, phase noise, etc. a typical degradation might incease the error by around 4-5 times, giving a conservative estimate for DoA resolution of about $\approx 3 - 4^\circ$.

7.6. Mechanical Design

The Pseudo-Doppler array is placed above the avionics bay where each antenna of the array connects to a SP6T RF switch. The array forms a circle of radius 39.837mm. The center antenna is connected to the rocket's LoRa packet radio. Due to physical proximity between the DoA array and the telemetry radio antenna, the DoA can only run when the telemetry radio isn't transmitting.

The size of the circular array also bounds the precision of the DoA estimation by:

$$\sigma_\theta \propto \frac{1}{r \cdot \mathrm{SNR} \cdot \sqrt{N}}$$

where r is the radius, N is the number of antennas, this gives an estimated resolution of ≈ 0.64, which is close to the previous unconservative estimate.

The recommendation for compact circular arrays is typically to have a radius of $0.1\lambda \leq r \leq 0.25\lambda$. In this case at a frequency of 915MHz, the radius is $\approx 0.121\lambda$.

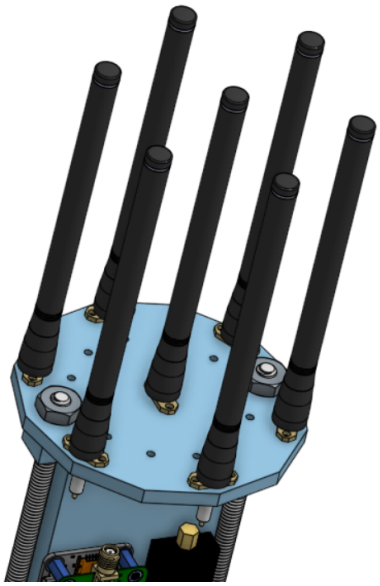


Fig. 7.3 Pseudo-Doppler array preliminary design in Avionics Bay.

7.7. Frequency Allocation at Competition

In order for the DoA system to work reliably, it requires a spectrally clean region around the carrier so that the induced Doppler tone can be measured without being buried in noise or modulation.

With a carrier at $915\mathrm{MHz}$, current estimates give a comfortable bandwitdth of $25\mathrm{KHz}$. This is contained within the LoRa spread spectrum bandwidth of $125\mathrm{Khz}$, so as long as this frequency allocation is respected, stray signals should not be a concern.

8. Avionics and Telemetry

8.1. Avionics Bay System and Wiring Overview

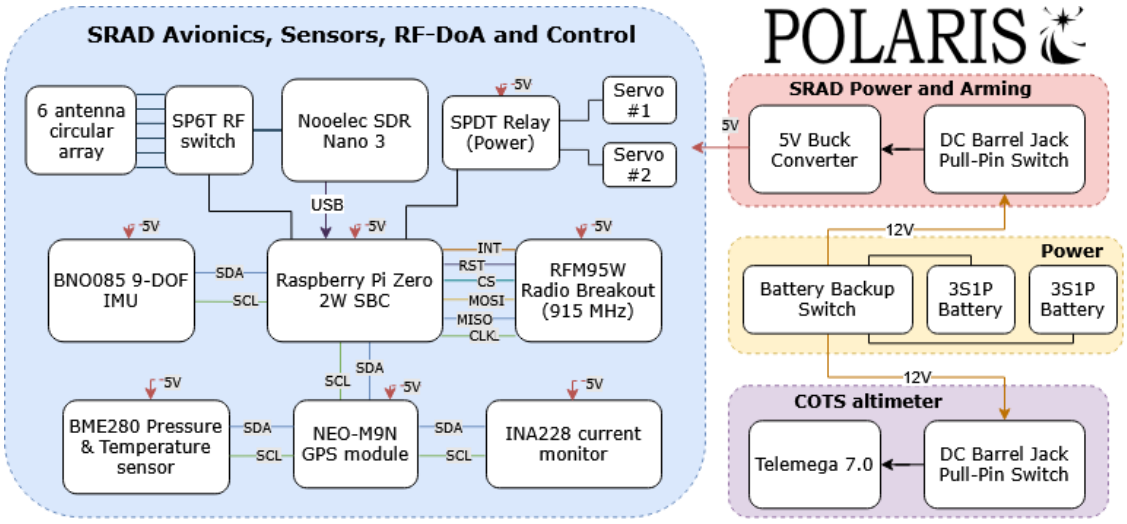


Fig. 8.1 Avionics & Telemetry system overview.

8.1.1. Controller

The *Polaris* SRAD avionics system's central logic is controlled by a Raspberry Pi Zero 2W single board computer (SBC), which allows for more complex peripherals and thread scheduling for the different state estimation and control systems on board.

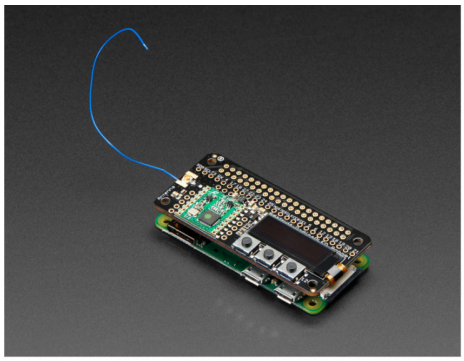


Fig. 8.2 Product image for RFM95W Radio Bonnet on Raspberry Pi Zero.

8.1.2. Interfaces

The Pi interfaces via USB with a Nooelec SDR Nano 3 for receiving the RF-DoA signal. The RFM95W LoRa transceiver is part of a Pi Zero compatible bonnet that includes three mechanical buttons as user interface and an OLED screen to be used for system diagnostics in pre-launch checks without needing to directly interface with a computer at all times.

The system also includes COTS breakout modules that interface using the SparkFun Qwiic modular I2C connector standard. This makes prototyping and assembly much easier by using a pre-made harness solution. The peripherals connected to this I2C bus are the BNO085 9-DOF sensor fused IMU, the BME280 pressure and temperature sensor, NEO-M9N GPS module, and INA228 voltage and current monitor.

8.2. Power Management

The avionics system is be powered by two 2200mAh 3S1P lithium polymer batteries connected to an automatic uninterruptible power controller to always use the highest voltage source. The 5V rail is supplied by a buck converter to which the servos is directly connected.

Testing the reliability of the battery backup switch will involve running the system on a typical mission load over an extended period while continuously monitoring the voltage ensuring there are no brownouts or power cycles during switching. This will also allow for better battery life estimates.

Table 8.1 Battery life estimate based on current draw

Load	Max Current (mA)
Raspberry Pi Zero 2W	600 mA
NESDR Nano v3	300 mA
Buck Converter	5 mA
Servos	2×300 mA
Sensors	100 mA
LoRa Packet Radio	50 mA
Total battery capacity	4400mAh
Battery life estimate	4.17h

8.3. Arming & Safing

Arming the avionics system is done by removing the DC barrel plug that closes the connection from the batteries to the power distribution system. Safing is done by re-inserting the plug into the jack.

8.4. Recovery Electronics

Since this rocket's recovery system relies solely on the motor's black powder charge, there is no backup recovery system. The COTS altimeter is the Telemega 7.0, which has a dual-deploy system if the option becomes necessary.

8.5. Telecommunication System

The telemetry system's main communication line is done using a half-duplex scheme with a RFM95W 915MHz LoRa transceiver on both the rocket and the ground.

Table 8.2 Half-duplex & RFDOA TX/RX loop

Source	**Rocket**		**Ground**	
Period	LoRa	RFDOA	LoRa	RFDOA
Rocket Telemetry	TX Packet	Inactive	RX	Inactive
Main DOA loop Rocket	RX	Active	TX Tone	Inactive
Main DOA Loop Ground	TX Tone	Inactive	RX	Active
	Repeat previous two steps N times			
Ground Station Command	RX	Inactive	TX Packet	Inactive

The Telemega 7.0 COTS flight controller uses the 433MHz band, which should not interfere with the RF-DoA system, but might overload the SDR receiver due to proximity. In the case that this becomes a problem, a 33 cm bandpass filter will be used to mitigate this issue.

8.5.1. Radio Link Budget

The link budget assumes two low-cost 915 MHz LoRa packet radios operating at 22 dBm with nominal receiver sensitivity, dipole antennas, and typical cable, connector, and an additional 'fudge factor' to account for other potential loss sources. Propagation is estimated using both free-space and conservative two-ray ground reflection models for a near-ground terrestrial link with low antenna heights. Additional unmodeled losses may come from partial Fresnel zone obstruction, antenna mismatch, ground conductivity, polarization mismatch, and environmental variability.

Link Budget Term	FSPL Model (dB)	2-Ray Model (dB)
Transmit Power	+22	+22
Transmit Antenna Gain	-2	-2
Transmit Cable / Connector Loss	-2	-2
Receive Antenna Gain	-2	-2
Receive Cable / Connector Loss	-2	-2
Implementation Loss	-3	-3
Path Loss	-111.7	-134.5
Received Power	-100.7	-123.5
Receiver Sensitivity	-130	-130
Link Margin	**29.3**	**6.5**

Table 8.3 Conservative link budget for two 22 dBm LoRa packet radios at 915 MHz over a 10 km terrestrial link using free-space and two-ray propagation models.

8.6. Electrical Ground Support Equipment

The ground station controller uses an identical antenna array to track the rocket using the same DoA scheme. Instead of using a Raspberry Pi SBC, the SDR is connected directly to the host ground station laptop. The ground station will also include a LoRa packet radio connected to a Teensy 4.1 microcontroller with its own GPS module.

8.6.1. Graphical User Interface (GUI)

The LoRa telemetry system handled by the ground station Teensy 4.0 microcontroller establishes a USB serial interface with the ground station laptop allowing it to relay all of the telemetry sent by the rocket, as well as local position from the GPS module. This information is then displayed in a custom graphical user interface written in Python using PyQt5. The interface also includes an embedded map that can be downloaded for the launch site in order to be used offline.

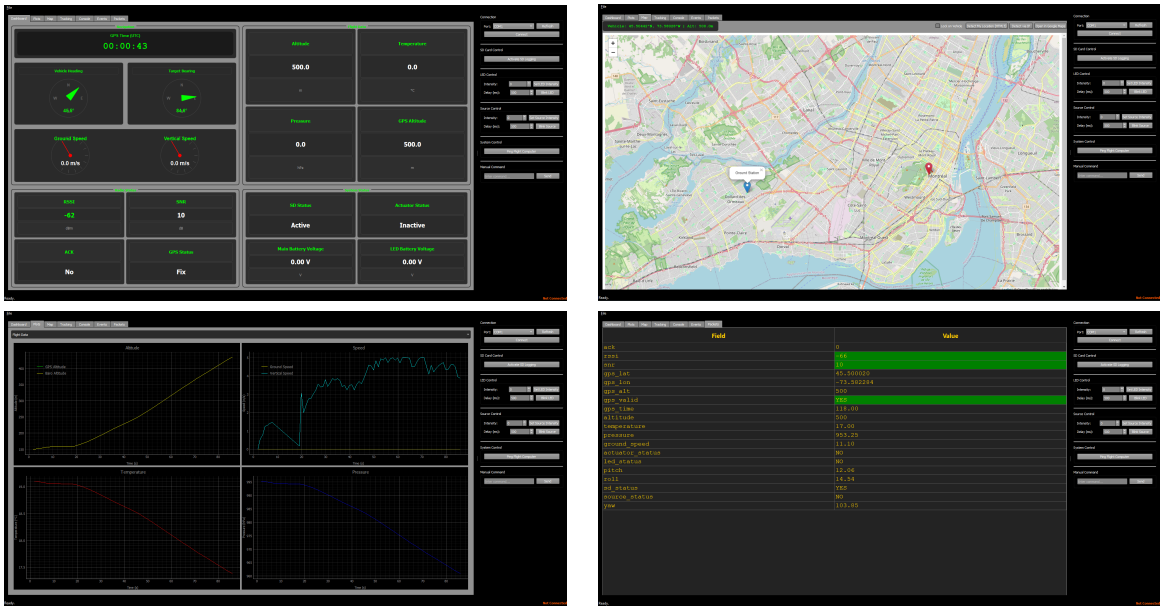


Fig. 8.3 Screenshots of graphical user interface panels with simulated sample data.

8.7. State Estimation

Table 8.4 Summary of Sensor Contributions to the State Estimator

Sensor	Primary Measurement	Constrained States
9-DOF IMU	Angular rate, specific force	Attitude, velocity (short-term)
Pseudo-Doppler RF	Direction of arrival (bearing)	Attitude, lateral position
Barometer	Pressure altitude	Vertical position
GPS	Position, velocity, time	Absolute position and velocity

The estimator is implemented as a loosely-coupled inertial navigation system with periodic measurement updaes. Inertial measurements propagate the system state forward in time:

$$\dot{\mathbf{p}} = \mathbf{v}, \quad \dot{\mathbf{v}} \approx R(\mathbf{a} - \mathbf{b_a}) + \mathbf{g}$$

while external sensors provide corrections to limit drift.

8.7.1. State Vector

The estimator maintains a compact state vector containing the platform's kinematic states and slowly varying sensor biases:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} & \mathbf{v} & \mathbf{q} & \mathbf{b}_g & \mathbf{b}_a & \mathbf{b}_p \end{bmatrix}^T$$

Table 8.5 State Vector Definition for the Navigation Filter

State	Symbol	Description
Position	$\mathbf{p}$	Platform position in the navigation frame
Velocity	$\mathbf{v}$	Platform velocity in the navigation frame
Attitude	$\mathbf{q}$	Body-to-navigation frame orientation
Gyro bias	$\mathbf{b}_g$	Slowly varying gyroscope bias
Accel bias	$\mathbf{b}_a$	Slowly varying accelerometer bias
Baro bias	b_p	Barometric altitude offset

Sensor biases are explicitly estimated to prevent long-term drift and to ensure consistency between inertial propagation and external measurements.

8.8. Roll Control Using RF-Based Target Pointing

The estimated state vector provides the attitude and angular rate information required to close a roll control loop for camera pointing. In particular, the attitude state $\mathbf{q}$ is used to determine the platform roll angle, while gyroscope measurements corrected by the estimated gyro bias $\mathbf{b}_g$ provide a low-latency estimate of roll rate. Together, these states form the feedback inputs to a PID roll controller.

The pseudo-Doppler direction-of-arrival (DOA) measurement supplies a bearing to the target RF emitter in the body frame. This bearing is converted into a desired roll angle that aligns the camera boresight with the estimated target direction. The roll control error is then defined as the difference between the desired roll angle inferred from the DOA measurement and the current estimated roll angle from the state vector.

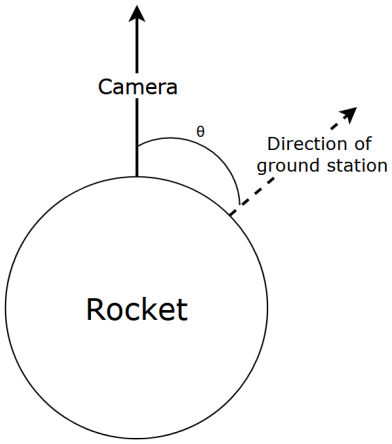


Fig. 8.4 Angle between on-board camera (zero angle) and direction of the RF source at the ground station (θ)

The PID controller generates a roll control command based on this error, its time derivative, and its accumulated history, allowing the system to reject disturbances while maintaining stable pointing. By relying on the state estimator rather than raw sensor measurements, the control system benefits from reduced noise, bias compensation, and improved robustness during dynamic flight conditions.

9. Roll Control System

9.1. Overview

In rocketry, roll corresponds to rotation around the vehicle's longitudinal axis. Although it does not directly alter the flight trajectory in the same way as pitch and yaw, uncontrolled roll can degrade overall stability, corrupt sensor data, and impose unnecessary structural loads. Therefore, a system that actively controls the roll attitude (angle and/or angular velocity) would be beneficial for a predictable flight.

This section is devoted to presenting the controller design and its simulations using a custom written dynamics model. The mechanical components are described in section 4..

9.2. Controller

This subsection first presents the dynamical model and some qualitative aspects of controller design. Then, a numerical example is given to demonstrate the design outcome. These numerical calculations and simulations are conducted using a smaller scale rocket model, before Iris was selected as the flight rocket. Therefore, numerical steps in this subsection should be viewed as a demonstration of the controller design process that will be implemented to design the actual flight controller.

9.2.1. Dynamics model

Developing a control system for a high-powered rocket requires a flight dynamics model and trajectory simulation tool beyond common open source packages such as OpenRocket. Therefore, attention is paid to first deriving six-degree-of-freedom (6-DOF) equations of motion. It is assumed that during the coasting phase—when the proposed controller is intended to operate—the rocket has constant mass, and structural deformation can reasonably be ignored. This gives gravity, aerodynamics drag, and normal force (aerodynamic lift) as the forces. The aerodynamic model is based on the Barrowman method [1]. These allow the use of a reduced but reasonably accurate nonlinear dynamics model.

The roll moments included are fin canting, aerodynamic damping, and the control actuation. Although the full dynamics couple all angular motions, the roll axis can be isolated for controller design by linearising the rotational equations about a nominal vertical flight condition. This linearisation results in a first-order transfer function from control moment to roll rate, enabling systematic frequency-domain controller design. The controller is then verified through simulation of the full nonlinear 6-DOF model.

9.2.2. Control objectives

During nominal ascent, the rocket must avoid accumulating an undesired angular velocity around its longitudinal axis. A non-zero roll rate ω_{b1}^{ba} continuously rotates the body frame, which:

- complicates the estimation of the aerodynamic angles,
- mixes the pitch and yaw axes due to body rotation,
- increases the structural loading on fins through periodic incidence variation,

For this reason, the primary objective is the regulation of

$$\omega_{b1}^{ba}(t) \to \omega_{b1,\mathrm{ref}}^{ba}(t).$$

In most flight phases, $\omega_{b1,\mathrm{ref}}^{ba} = 0$. Different roll rates (e.g. 30°/s) are used for verifying tracking performance later.

The controller is therefore designed to satisfy:

- fast roll-rate regulation,
- overshoot $< 10\%$,
- settling time ~ 1 s,
- phase margin $> 50^\circ$,
- closed-loop bandwidth $\omega_c \simeq 12\ \mathrm{rad/s}$.

9.2.3. PI controller structure

A PI controller is preferred because:

- it enforces zero steady-state tracking error in roll rate,
- it is easy to implement onboard,
- and does not amplify measurement noise as a derivative term would.

In Laplace variables:

$$C_p(s) = K_{p,p} + \frac{K_{i,p}}{s} = K_{p,p}\Big(1 + \frac{1}{T_i s}\Big), \qquad T_i = \frac{K_{p,p}}{K_{i,p}}.$$

The control moment around the roll axis is

$$m_{b1}^{\mathcal{B}c,roll}(t) = K_{p,p}\, e_p(t) + K_{i,p} \int_0^t e_p(\tau)\, d\tau,$$

where

$$e_p(t) = \omega_{b1,\mathrm{ref}}^{ba}(t) - \omega_{b1}^{ba}(t)$$

is the difference between reference and actual roll rate.

9.2.4. Frequency domain tuning

From the linearised roll dynamics equations, a transfer function, $G_p(s)$, can be derived. Numerical values can be obtained by substituting in aerodynamic coefficients. For now, choosing $U = 50\ \mathrm{m/s}$ gives:

$$G_p(s) = \frac{50}{s + 0.623}.$$

Choosing $\omega_c = 12\ \mathrm{rad/s}$, $\zeta_c = \sqrt{2}/4 \simeq 2.4\ \mathrm{rad/s}$, the PI gains are then set so that

$$|C_p(j\omega_c) G_p(j\omega_c)| = 1$$

The MATLAB tuning yields

$$K_{p,p} = 0.236, \qquad K_{i,p} = 0.566.$$

The figures below show some of the characteristics of the controller.

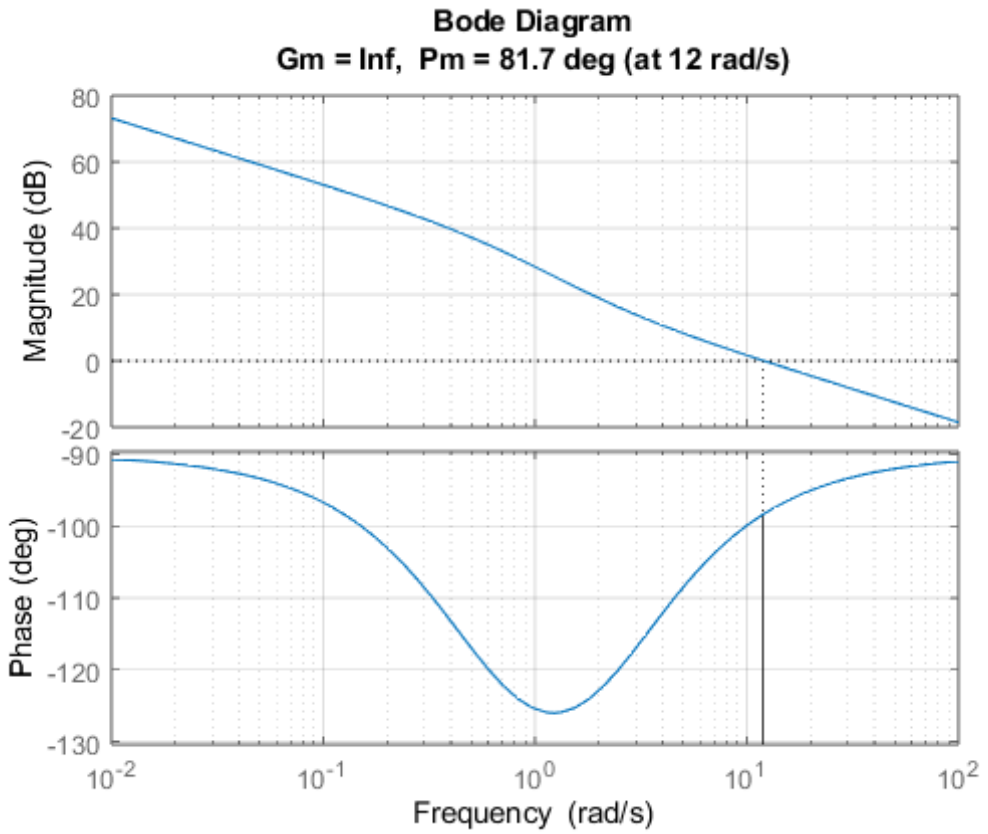


Fig. 9.1 Open-loop Bode diagram of $L(s) = C_p(s)G_p(s)$. The phase margin is approximately $82°$, indicating a highly robust roll channel. The crossover occurs at $12\ \mathrm{rad/s}$, as imposed during tuning.

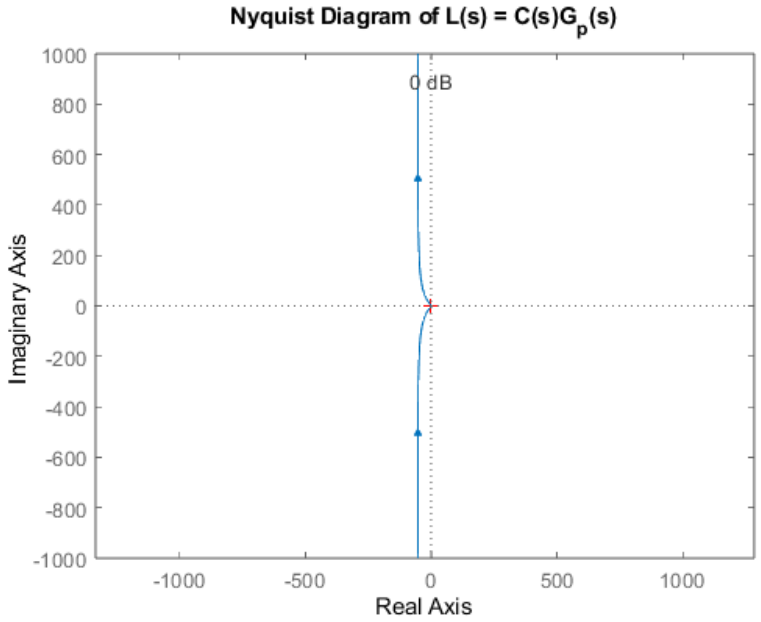


Fig. 9.2 Nyquist plot of the roll open loop. The curve does not encircle the point $(-1, 0j)$, confirming the stability. Its large separation from the $-180°$ axis indicates excellent gain robustness.

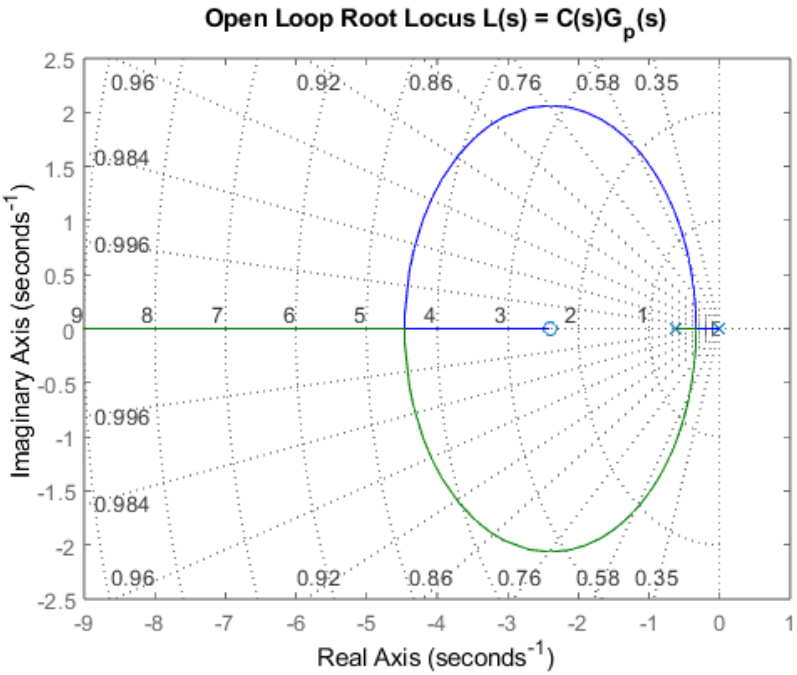


Fig. 9.3 Root locus of the open-loop roll dynamics. The closed-loop pole moves along a real/complex branch and remains in the left-half plane for all positive gains, confirming the absence of hidden instabilities.

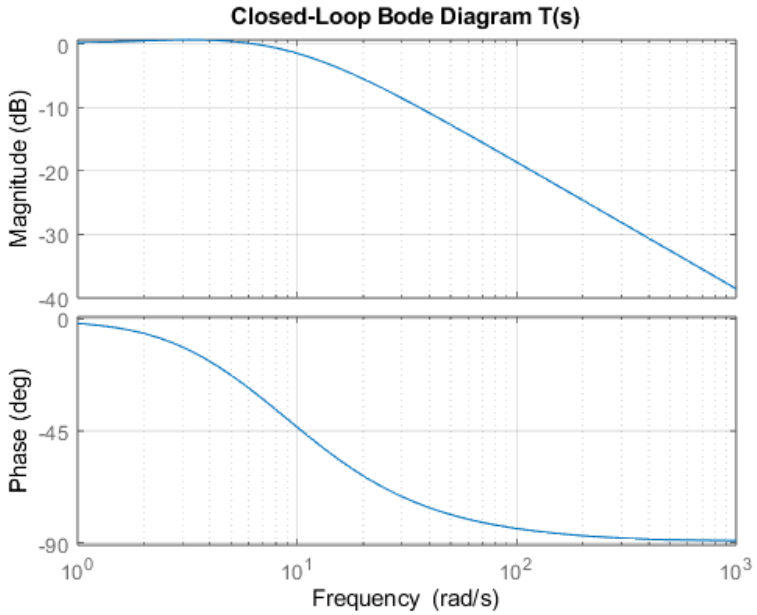


Fig. 9.4 Closed-loop transfer function $T(s) = \Omega_{b1}^{ba}(s)/\Omega_{b1,\mathrm{ref}}^{ba}(s)$. The magnitude remains flat at low frequency (good tracking), with a bandwidth near $12\ \mathrm{rad/s}$. The phase approaches -90° at high frequency, as expected for a first-order system.

The controller achieves:

- fast, well-damped roll-rate tracking,
- strong robustness margins,
- excellent disturbance rejection,
- zero steady-state error.

9.2.5. Simulation Results

The nonlinear 6-DOF simulation is performed using with a fixed time horizon and an event function detecting the apogee. The full aerodynamic model, thrust curve, and gravity are included. The roll PI controller is activated after motor burnout, using the control law

$$m_{b1}^{\mathcal{B}c}(t) = K_{p,p}(p_{\mathrm{ref}} - \omega_{b1}^{ba}) + K_{i,p}\,\xi(t), \qquad \dot{\xi}(t) = p_{\mathrm{ref}} - \omega_{b1}^{ba}.$$

Two types of simulations are presented:

- response to reference changes and disturbances,
- a full-fidelity free-flight simulation with $p_{\mathrm{ref}} = 0$, representative of nominal rocket flight.

9.2.5..1 Response to reference changes and disturbances

1. Reference tracking and disturbance rejection Step responses for roll-rate commands are simulated. Fig. 9.5 shows the roll-rate response for three reference commands $p_{\mathrm{ref}} = 10, 30, 60\ \mathrm{deg/s}$. For each case, the PI controller produces

- a fast rise time (less than 1 s),
- a small overshoot (that increases with the command amplitude, but stays under 10%),
- convergence to the requested steady-state roll rate.

Nonlinear effects become visible for the highest reference command ($60^\circ/\mathrm{s}$), where aerodynamic damping increases strongly with p.

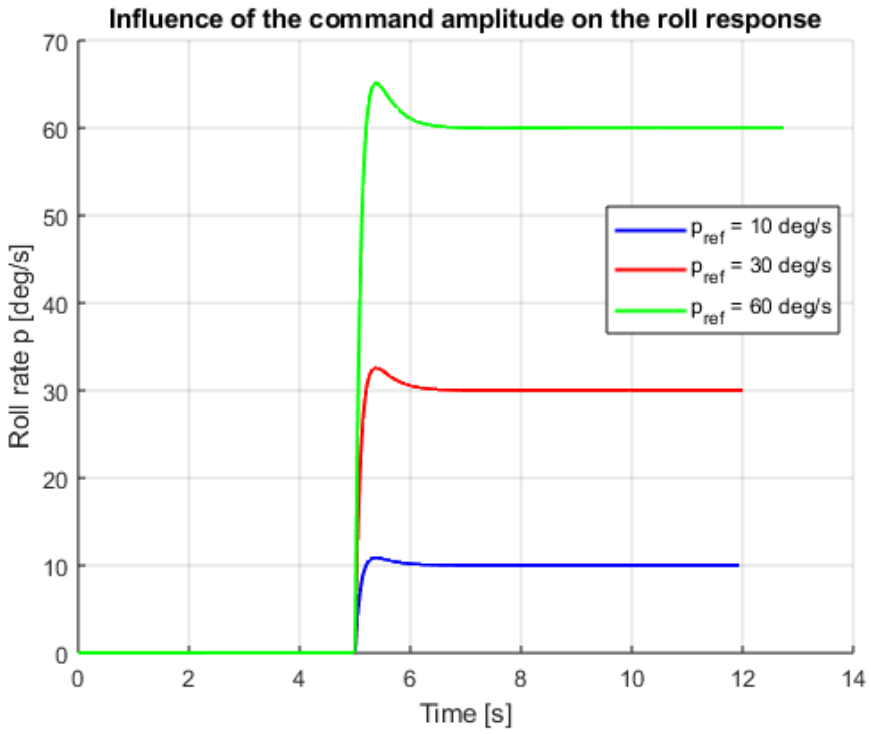


Fig. 9.5 Roll-rate response for three roll-rate commands p_{ref}.

2. Impulsive roll disturbance rejection Figure 9.6 shows the response to a short impulsive aerodynamic torque applied during coasting. Without control, the roll rate diverges freely (open loop), whereas the PI controller rapidly brings it back to the reference. This confirms that the closed loop effectively removes roll disturbances, augmenting roll stability.

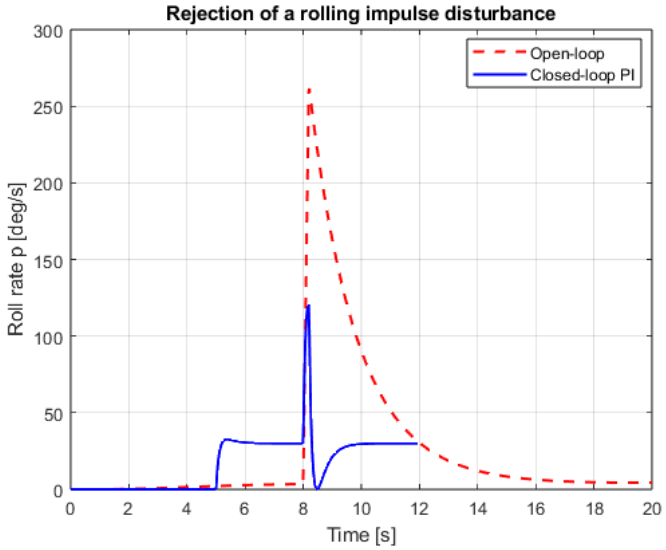


Fig. 9.6 Rejection of an impulsive roll disturbance: open-loop vs closed-loop.

9.2.5..2 Full-fidelity simulation with $p_{\mathrm{ref}} = 0$

1. Angular velocities and Euler angles Figure 9.7 shows the evolution of angular velocities and Euler angles: ϕ, θ and ψ for roll, pitch and yaw, respectively. The PI controller maintains the roll rate near zero (blue curve) despite nonlinear aerodynamic couplings. Pitch and yaw are only indirectly affected, confirming that the roll controller does not destabilise the other axes.

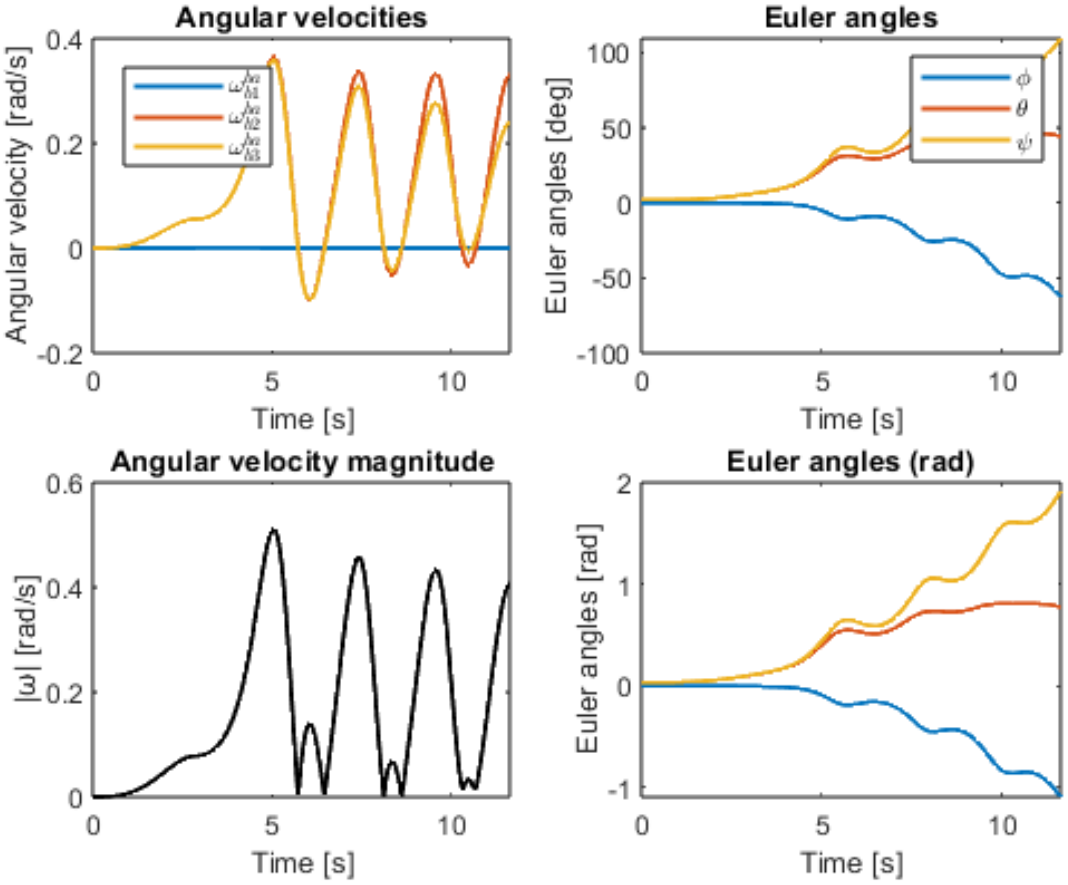


Fig. 9.7 Angular velocities and Euler angles during the controlled flight ($p_{\text{ref}} = 0$).

2. Control moment evolution Figure 9.8 displays the roll control moment $m_{b1}^{\mathcal{B}c}(t)$. Its amplitude remains extremely small (of order 10^{-3} to 10^{-4} $\mathrm{N} \cdot \mathrm{m}$), well within actuator capacity. This confirms that maintaining $p \approx 0$ requires minimal control effort thanks to natural aerodynamic damping.

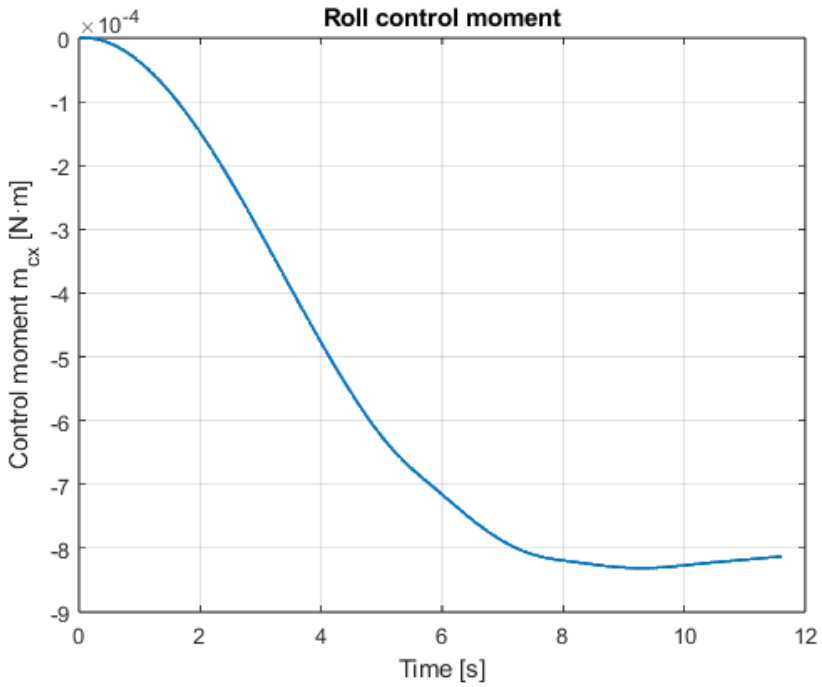


Fig. 9.8 Roll control moment $m_{b1}^{\mathcal{B}c}(t)$.

3. Integral state evolution Figure 9.9 shows the internal integral state $\xi(t)$. It converges smoothly and remains bounded, confirming the absence of integral windup and the overall good conditioning of the PI loop.

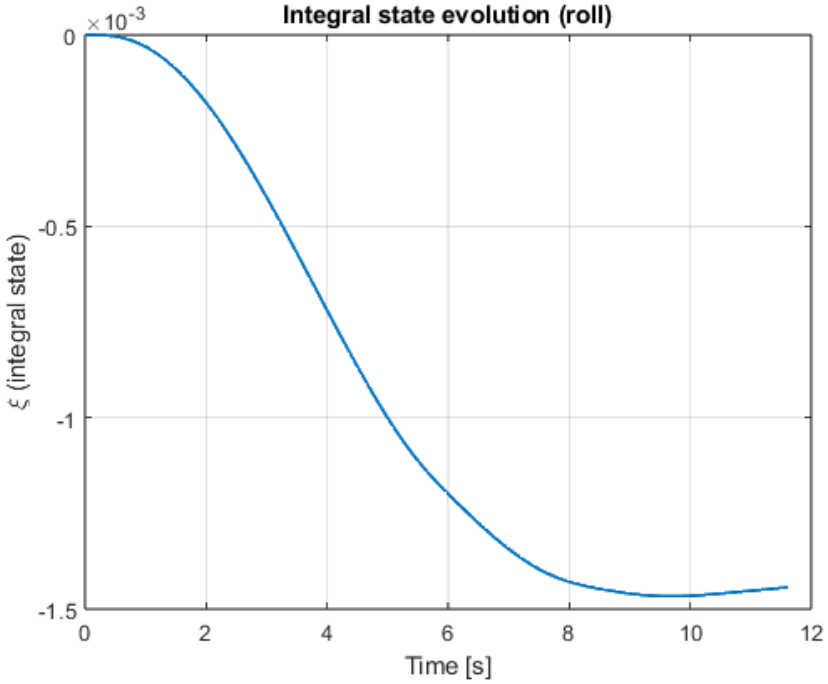


Fig. 9.9 Integral state evolution $\xi(t)$ for roll control.

4. Roll rate tracking Figure 9.10 compares open-loop and closed-loop roll-rate responses. In open loop, the rocket slowly spins up due to aerodynamic asymmetry; closed loop keeps the roll rate nearly zero at all times.

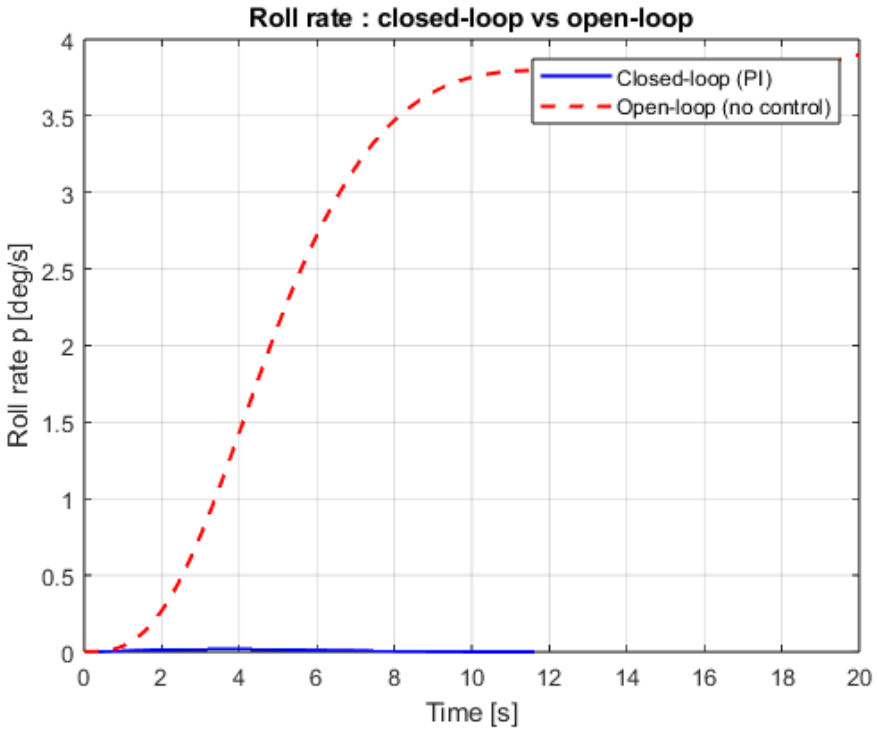


Fig. 9.10 Open-loop vs closed-loop roll-rate response.

5. Tracking error Figure 9.11 shows the tracking error $e_p = p_{\text{ref}} - p$. The error remains extremely small after a short transient, confirming excellent roll stabilisation.

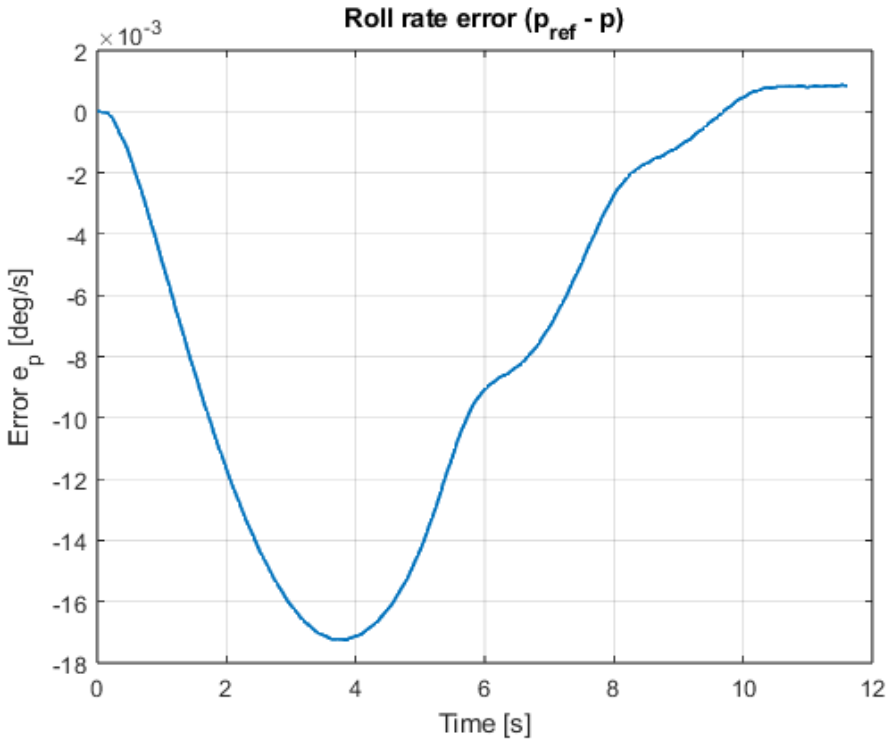


Fig. 9.11 Roll-rate tracking error for $p_{\mathrm{ref}} = 0$.

6. Zoom on transient roll control Figure 9.12 presents a zoom around lift-off. The controller quickly removes the small initial rotational motion, reducing error to nearly zero within less than 1 second.

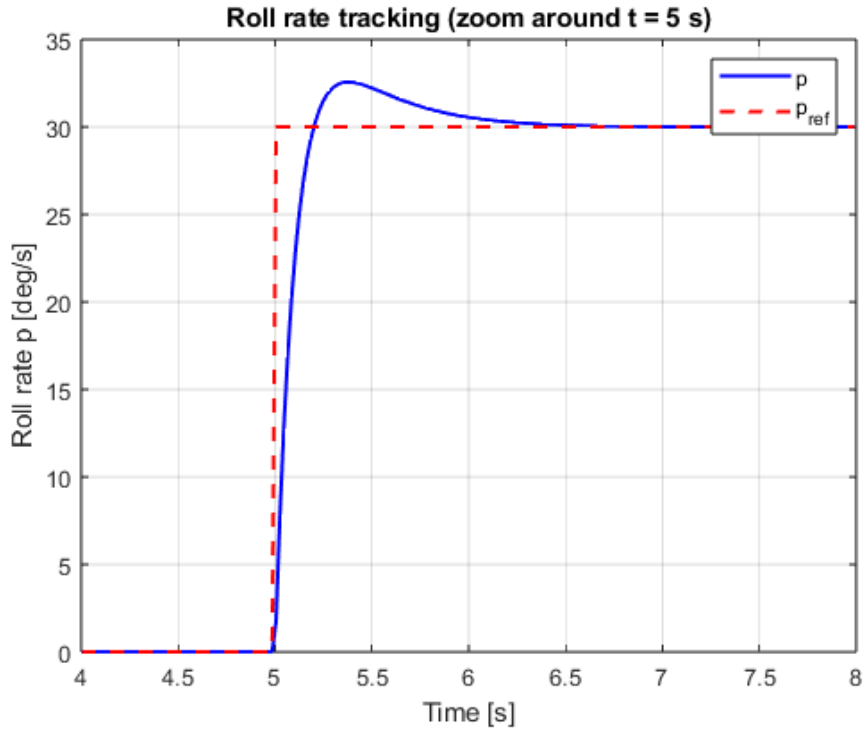


Fig. 9.12 Zoom on the roll transient at the beginning of the flight.

9.2.6. Subsection Summary

Nonlinear simulations confirm that the roll PI controller designed from the linearised model performs as intended on the full 6-DOF dynamics. The closed loop tracks roll-rate commands with fast, well-damped responses, rejects impulsive disturbances during coasting, and maintains essentially zero roll rate in nominal conditions with very modest control effort. The overall translational trajectory remains virtually unchanged, which means that roll control primarily acts on body orientation with little degradation on the ballistic performance of the rocket.

10. Programmatic Risks, Failure Modes and Effects Analysis, and Mitigation Strategies

Table 10.1 Hazards level matrix.

		Likelihood				
		Near certain (5)	Probable (4)	Possible (3)	Improbable (2)	Rare (1)
Criticality	Catastrophic(5)	10	9	8	7	6
	Critical (4)	9	8	7	6	5
	Moderate (3)	8	7	6	5	4
	Not critical (2)	7	6	5	4	3
	Negligible (1)	6	5	4	3	2

Table 10.2 Risk priority and mitigation.

Hazard rating	Action
Extreme (8-10)	Detailed plan of action required. Need immediate action to mitigate the risk. All related activities needs to be put hold until the hazard is eliminated or reduced to a lower level.
High (6-7)	Detailed plan of action required. Action should be taken as soon as a mitigation strategy is found.
Medium (5)	Action should preferably be taken to mitigate the risk, but not required. Monitoring of the risk is necessary.
Low (2-4)	No significant hazard exists. Simple monitoring of the risk is sufficient.

10.1. Programmatic Risks

Table 10.3 Programmatic risk assessment.

Code	Risk	Likelihood (1-5)	Criticality (1-5)	Level (1-10)	Impact	Mitigation strategy
Overall						
OV1	Overbudget	3	4	7	Unable to manufacture all components.	Design for minimizing cost; looking for sponsors
Controls						
CS1	No suitable small scale test launch location	4	3	7	Unable to validate controller performance through flight test	Design for high robustness; exploring ground testing methods
Avionics						
AV1	Long lead times on critical electronics components.	2	2	4	Unable to purchase and assemble PCBs for SRAD module	Purchase early and enough components to last through all design revisions; search for alternative components. Design around components with compatible alternatives (drop-in replacements).

10.2. Failure Mode & Effect Analyses

Table 10.4 Failure modes and effects analysis (FMEA) by subsystems.

Code	Component	Potential failure mode	Likelihood (1-5)	Criticality (1-5)	Potential causes	Recommended actions
				Controls		
CS1	Control law	Incorrectly tuned controller	2	5	Incorrect assumptions in flight dynamic model; erroneous aerodynamic coefficients	Design the controller for a high robustness; flight test on small scale rockets if possible
CS2	Control surface	In-flight structural failure from aerodynamic loads.	1	5	Insufficient structural integrity, manufacturing defects.	Select actuators and control surface materials for a high factor of safety
CS3	Subsystem level	Safety lock disengage during powered flight and/or failure to reengage after any of ”safe lock” conditions from DTEG has been met.	2	4	Defects in mechanism design and/or defects in flight computer logic	Extensive ground testing covering off-nominal conditions (flight trajectories, load cases, electric failure, etc.)
				Avionics		
AV1	RF-DoA	Direction finding system cannot reliably lock on the signal target or see large discrepancies in angle, making the control surfaces perform large erratic sweeps	1	4	Multi-path effects create correlated secondary signals in the antenna array, as well as other nearby signals emitted by other teams.	Confidence in the measured direction of arrival must be quantified, and a threshold confidence level must be achieved before attempting a maneuver. Additionally, the estimator must filter out erratic jumps in measured angle. The exact bandwidth needed for reliable DoA will be determined in order to reserve it with Launch Canada.
AV2	Tracking	GPS fails to lock or loses lock during flight/recovery	3	5	GPS antenna is misplaced in airframe or GPS module enters soft locked state due to high acceleration	Ensure the GPS antenna has proper visibility of the sky at all times and configure the U-Blox module to allow higher accelerations. The RF-DoA system also allows for a secondary tracking mode for recovery
AV3	Tracking	Radio signals sent by rocket are too weak during flight	1	3	Antenna and cable losses are too significant and environmental factors such as multipathing cause significant interference	Ensure that the antennas both on the rocket and ground have good omnidirectional gain patterns ensuring clear coverage.
AV4			1	4		

Code	Component	Potential failure mode	Likelihood (1-5)	Criticality (1-5)	Potential causes	Recommended actions
AV5			1	4		

References

[1] J. S. Barrowman, The practical calculation of the aerodynamic characteristics of slender finned vehicles, m.Sc. thesis (1967).